ERC Freshie Oreo Idea

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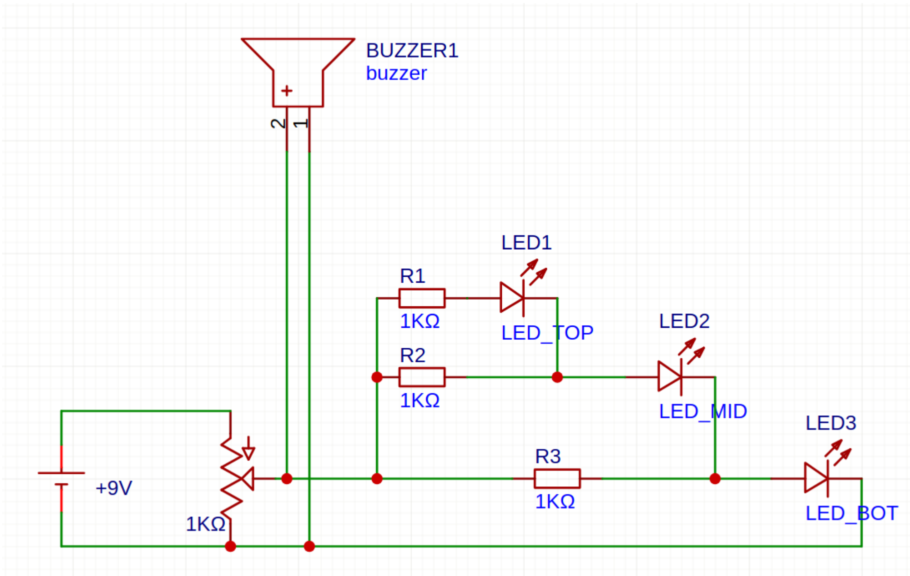
1 Introduction

Idea: Potentiometer to make a sort of danger level indicator

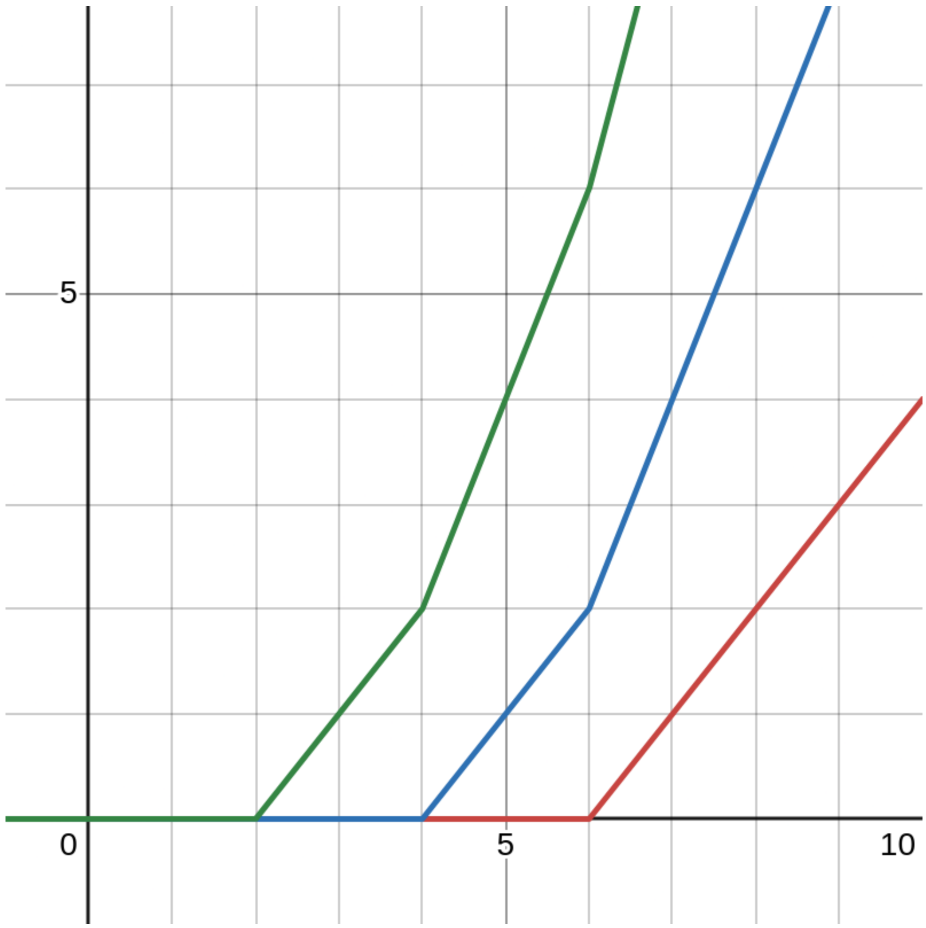
- A circuit with 3 LEDs and a buzzer, which gives an output of different LEDs lighting up and a buzzer that varies with the rotation of the potentiometer.
- The more the potentiometer is turned, the buzzer buzzes more and the LEDs glow brighter, where the bottom always glows brighter than the top.
- When the voltage increases, the bottom LEDs glow brighter and the top LED switches on, so it should look like some kind of meter filling up as the top LEDs slowly glow up and fill while you turn the potentiometer.
- The circuit is easy to understand and there are no complicated components (I believe in our oreo, there was an op amp and I got very confused about why it's there and what it does). You could use op amps here as well to make the LED meter a lot better using a comparator, but then the freshies won't be able to understand it, which makes it bad.

2 Schematic



- Buzzer doesn't require resistance because its resistance is already very high
- Potentiometer is 1K so that it is decently high to prevent too much power usage, but low enough such that the resistance LED part of the circuit isn't much lower, to prevent the parallel resistance making the potentiometer very sensitive in one region and not sensitive in the other region.
- Top LED always glows less brighter than middle which glows less brighter than bottom
- If voltage across LED section is V, and forward bias voltage drop across each LED is 2V:
  - Current through TOP LED:
    - (V - 6) / 1K if $6 < V$
    - 0 if $V < 6$
  - Current through MID LED:
    - (V - 4) / 1K + (V - 6) / 1K = (2V - 10) / 1K if $6 < V$
    - (V - 4) / 1K if $4 < V < 6$
    - 0 if $V < 4$
  - Current through BOT LED:
    - (V - 2) / 1K + (V - 4) / 1K + (V - 6) / 1K = (3V - 12) / 1K if $6 < V$

    * (V - 2) / 1K + (V - 4) / 1K = (2V - 6) / 1K if $4 < V < 6$
    * (V - 2) / 1K if $2 < V < 4$
    * 0 if $V < 2$
  - Note: Current never crosses 20mA (Max safe current through LED)
  - Graph: https://www.desmos.com/calculator/m5wlm2gdkd



3 Bill of Materials

https://docs.google.com/spreadsheets/d/1of8mBkQk77rY1RWmuujmq0QEFTayatkbsGKXgm0VQ5w/edit?usp=sharing

4 Sources of Information

- Limiting resistance for LEDs:
  https://www.gsnetwork.com/led-resistor-values-for-current-limiting-resistor

- Active vs Passive buzzer: https://www.youtube.com/watch?v=goUNElFt9Jg